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Chiang

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(54) **SPINAL LAMINA PROTECTOR**

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A61B 17/70 (2006.01)

(52) **U.S. Cl.**
CPC ***A61B 17/7071*** (2013.01); ***A61B 17/707*** (2013.01)

(58) **Field of Classification Search**
CPC A61B 17/707; A61B 17/7071; A61B 17/8004; A61B 17/8023
See application file for complete search history.

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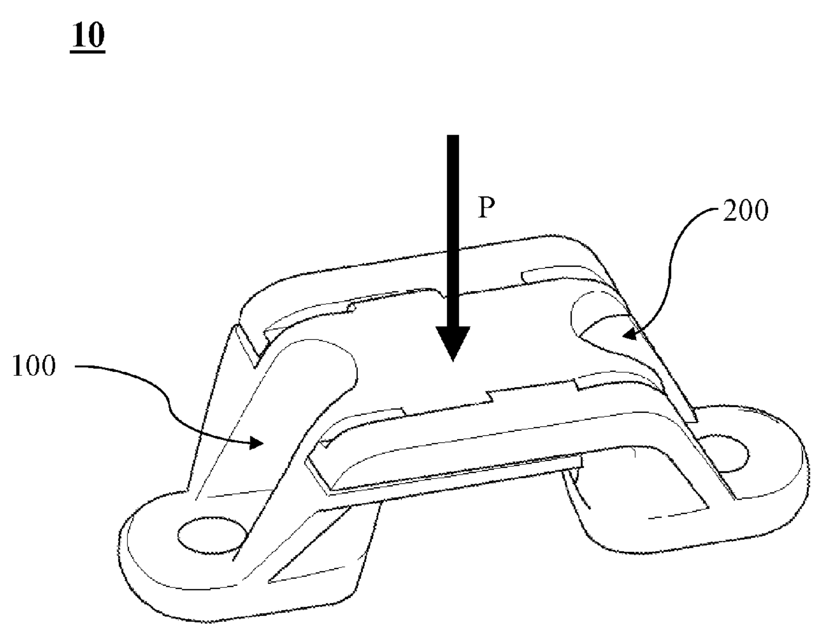
(57) **ABSTRACT**

The invention discloses a spinal lamina protector. The spinal lamina protector comprises a first protection part and a second protection part. Specifically, the first protection part comprises a first engaging portion, a first holding portion and a supporting portion, and the supporting portion and the first holding portion form an engaging rail. The engaging rail makes a second engaging portion of the second protection part insert in, and a second holding portion of the second protection part holds the first engaging portion while the second engaging portion is in the engaging rail. Therefore, the abovementioned structure of the spinal lamina protector forms a stable and length-adjustable structure, even if a pressure is pressed on the spinal lamina protector.

10 Claims, 9 Drawing Sheets



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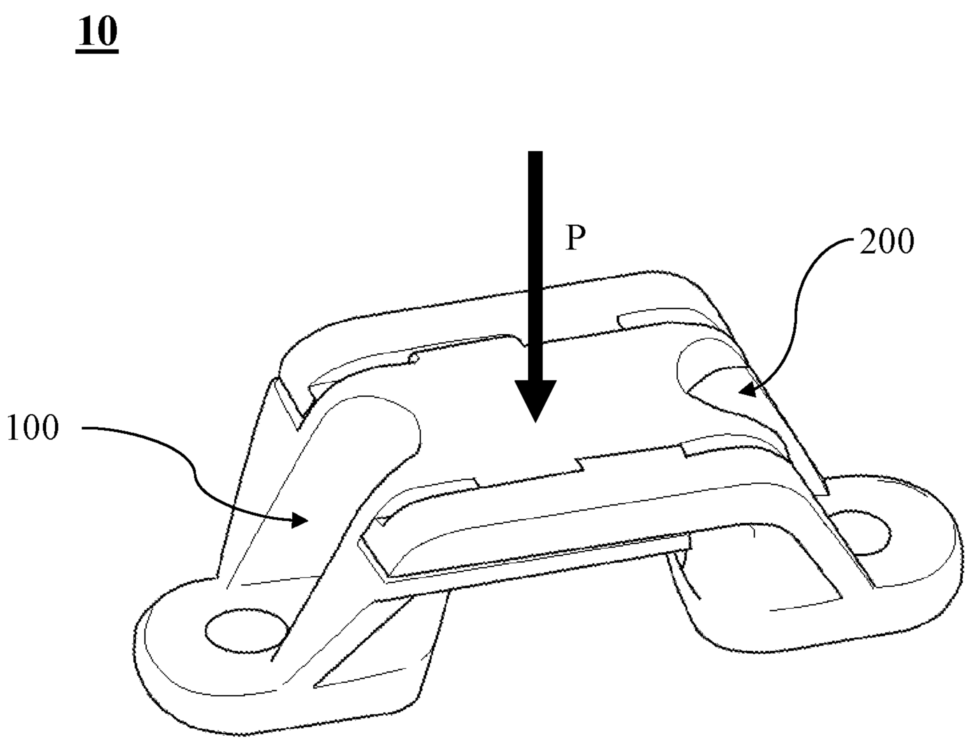

Fig. 1

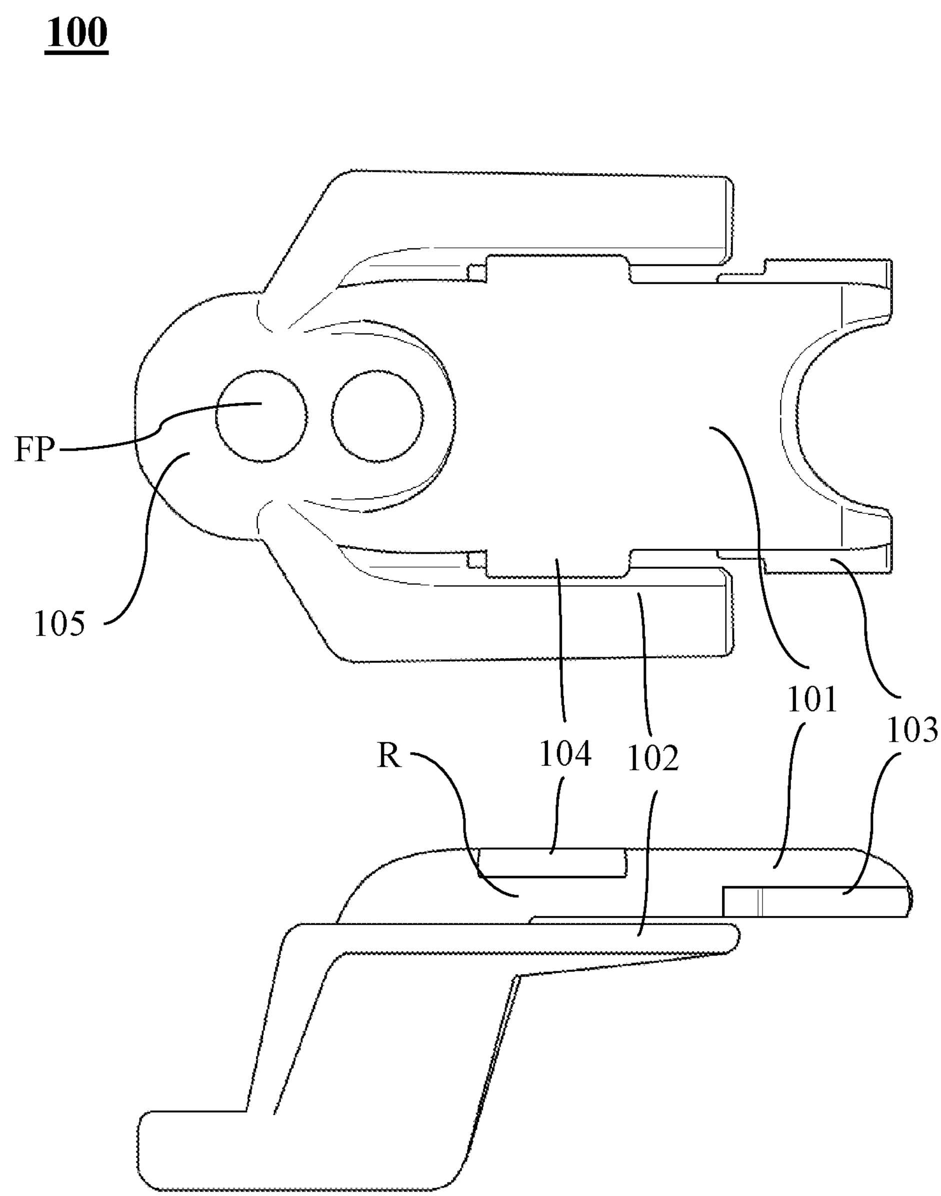

Fig. 2

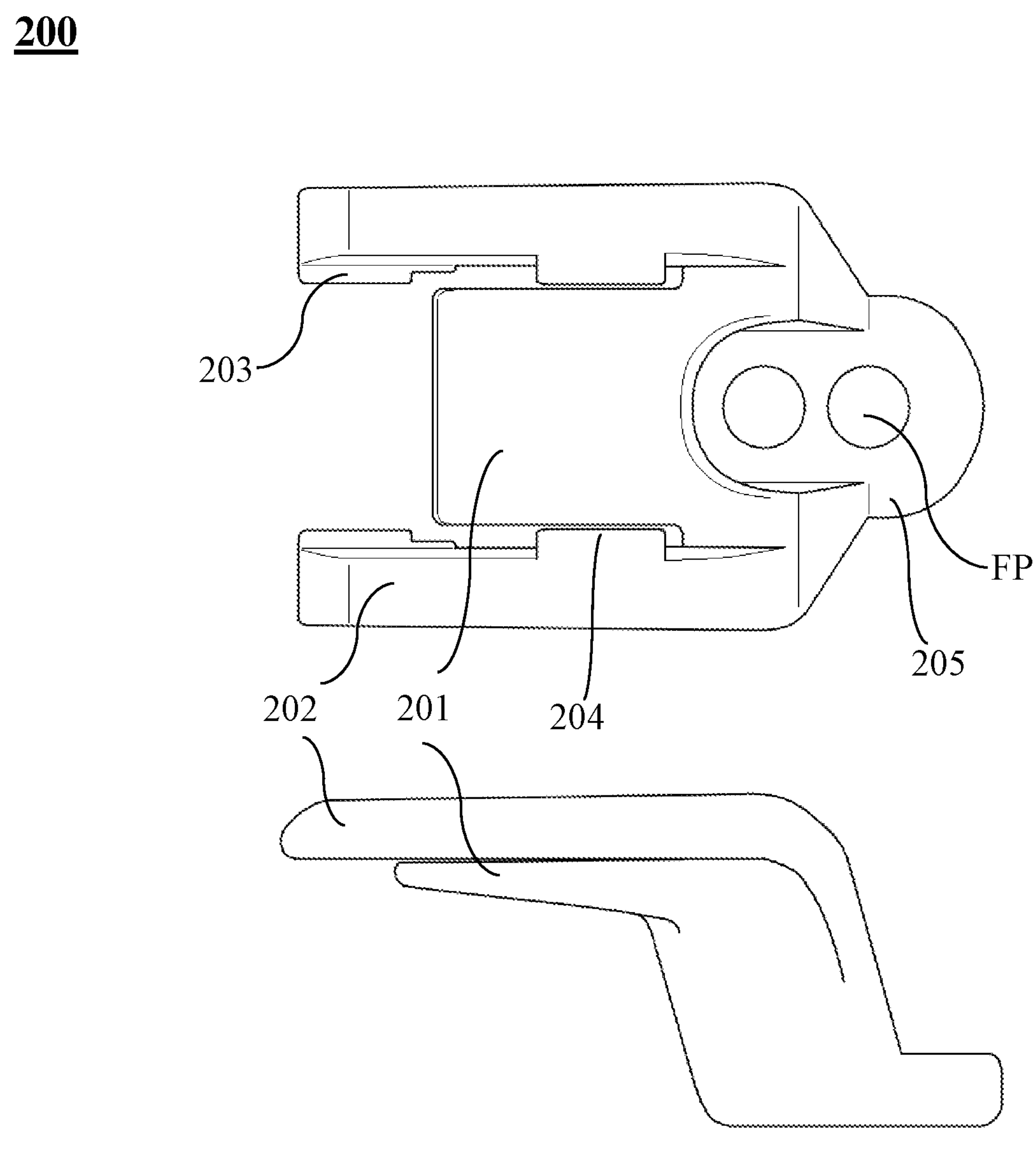

Fig. 3

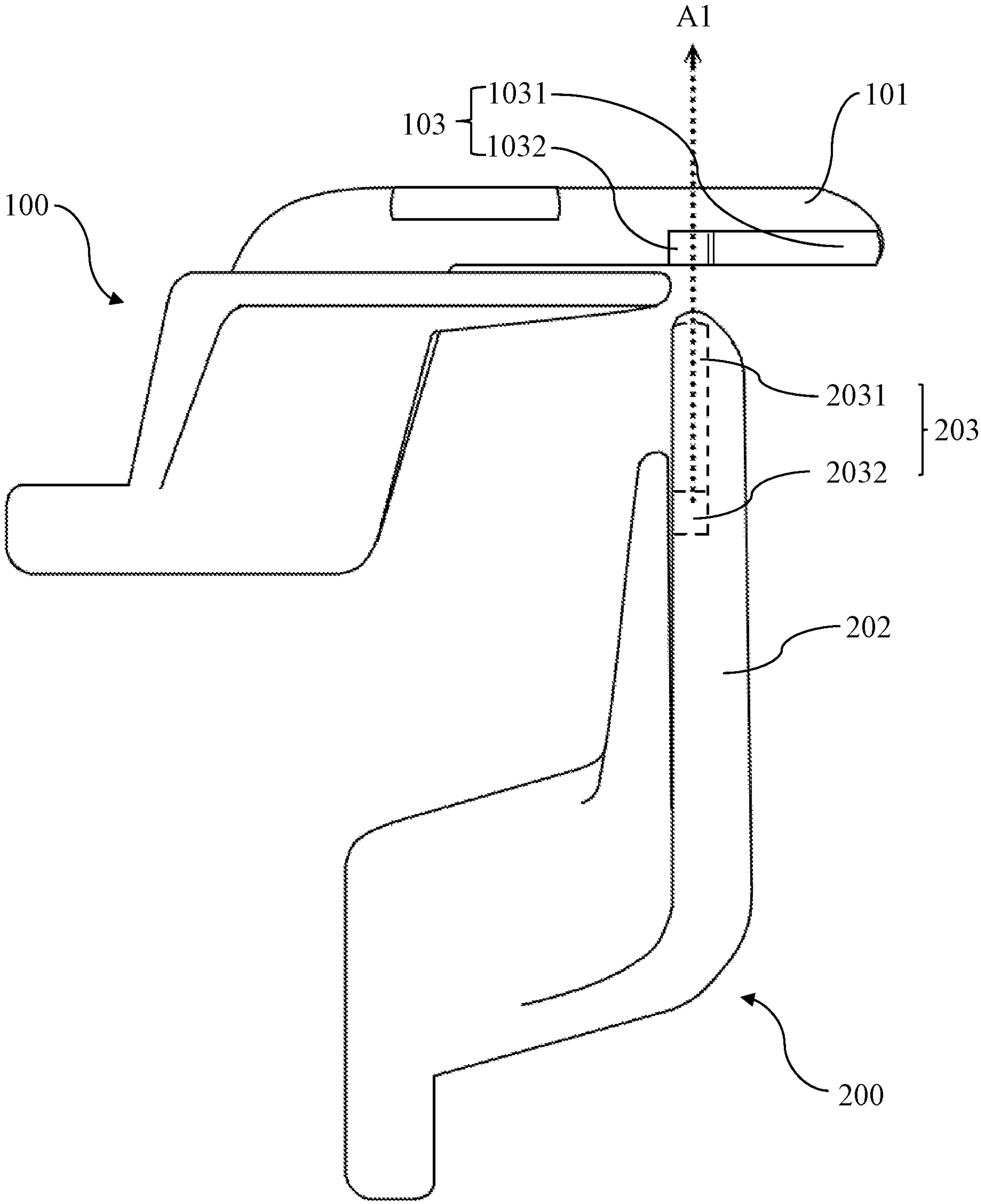

Fig. 4

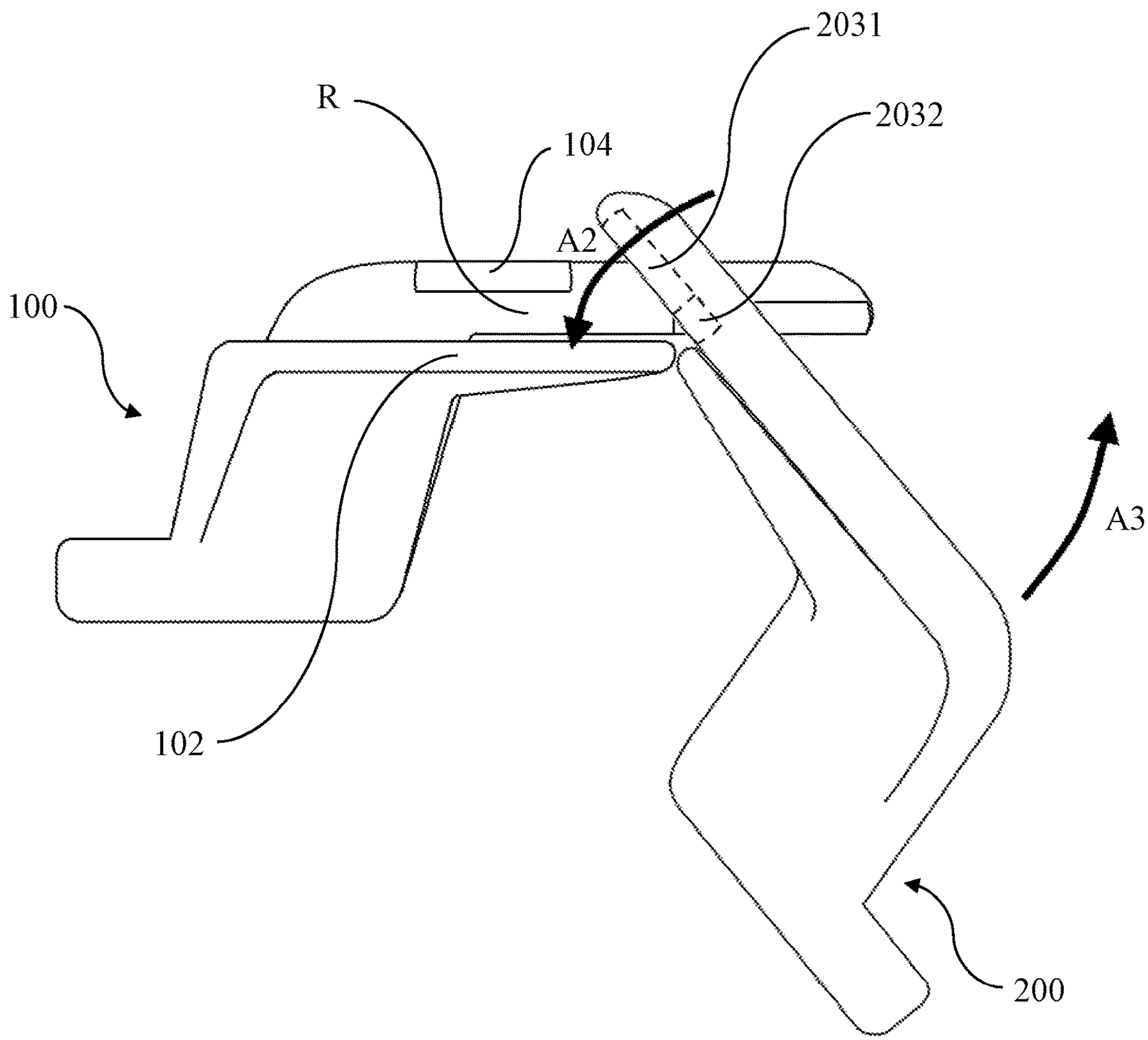

Fig. 5

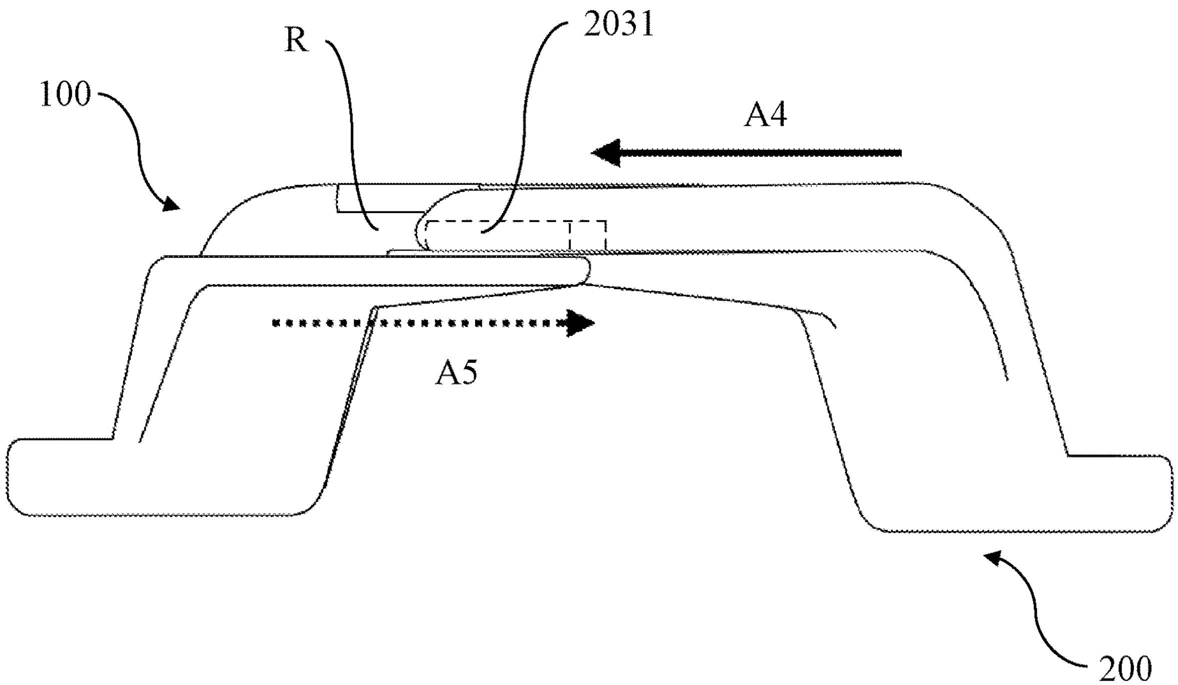

Fig. 6

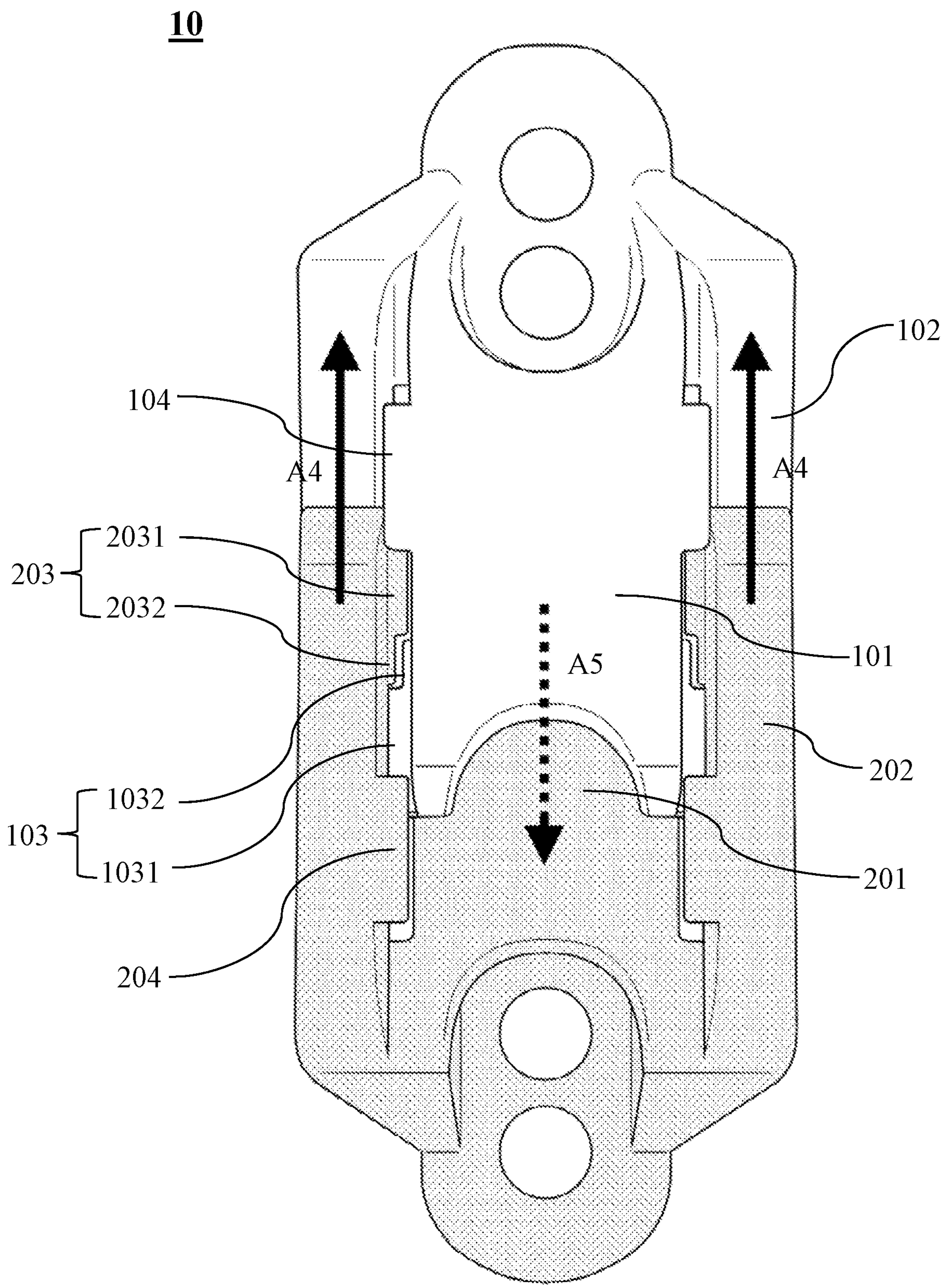

Fig. 7

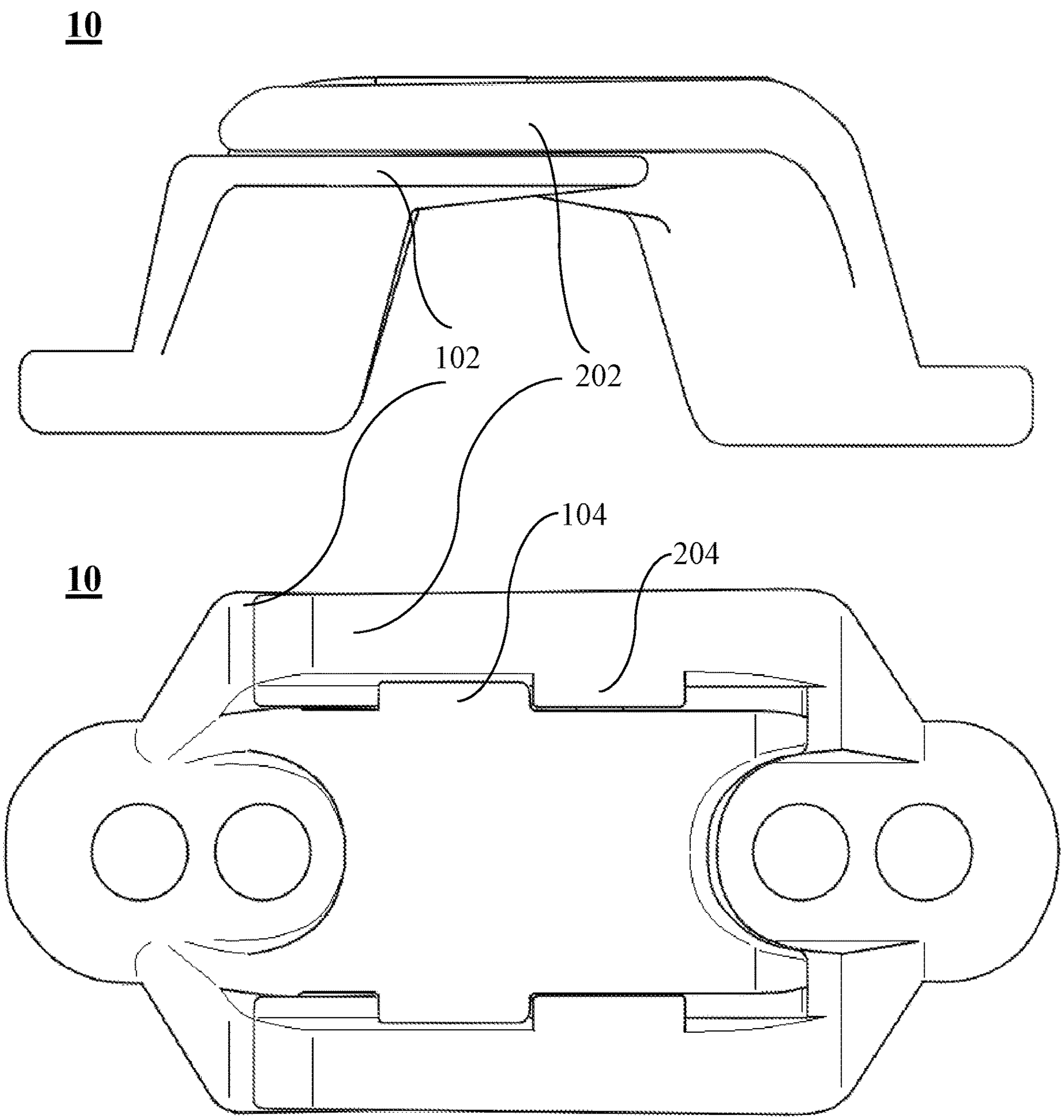

Fig. 8

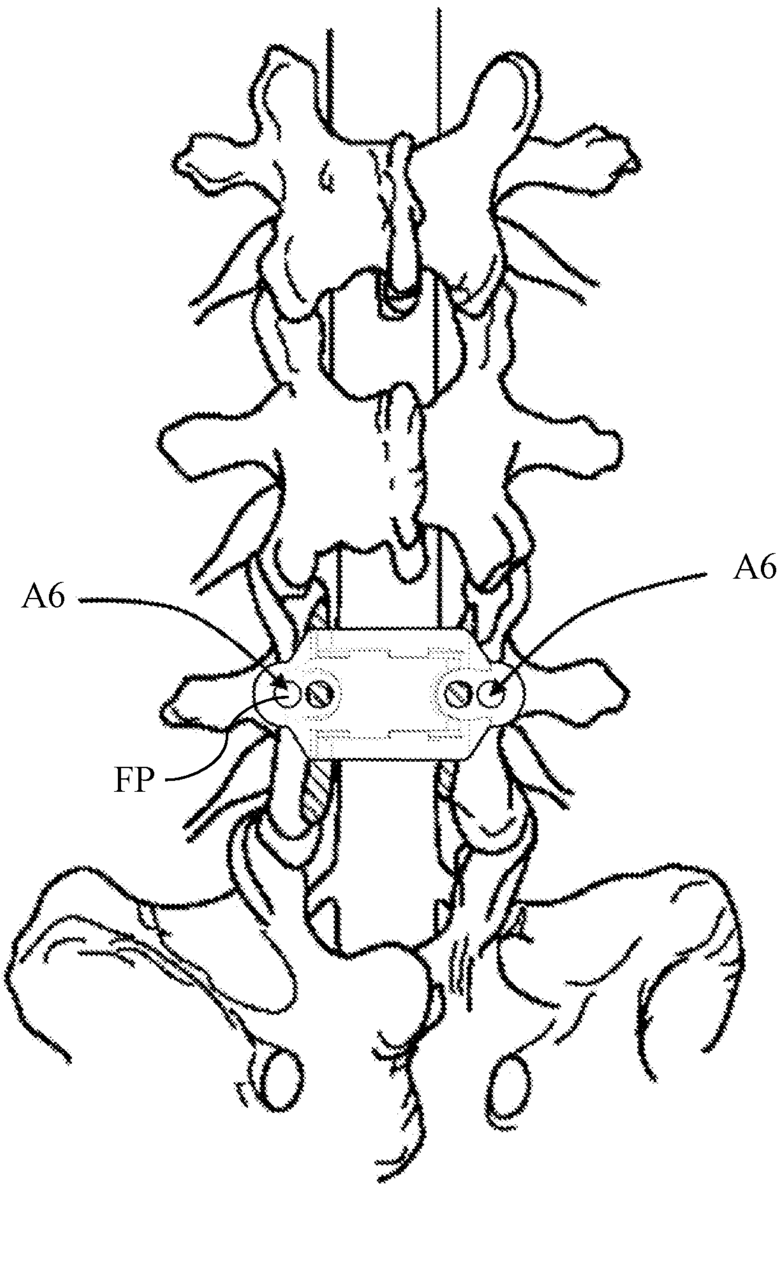

Fig. 9

SPINAL LAMINA PROTECTOR

TECHNICAL FIELD

The present invention relates to a spinal lamina protector which is designed to perform a stable and length-adjustable structure and for protecting the spinal cord from adhesion and external pressure after implanted in human body.

BACKGROUND OF RELATED ARTS

The lamina is a posterior arch of the vertebral bone lying between the spinous process and the lateral pedicles with the transverse processes of each vertebra. The pair of laminae, along with the spinous process, make up the posterior wall of the bony spinal canal. However, some diseases may cause the bony spinal canal narrowed and pressing the spinal nerves, e.g. spinal stenosis.

Most commonly, a laminectomy is performed to treat spinal stenosis. Spinal stenosis is the single most common disease that leads to spinal surgery, of which a laminectomy represents one component. The lamina of the vertebra is removed or trimmed to widen the spinal canal and create more space for the spinal nerves and thecal sac.

A laminectomy can treat severe spinal stenosis by relieving pressure on the spinal cord. After this surgery, part of the spinal cord which is originally protected by the lamina may directly contact to the surrounding tissue due to the laminae per se have been removed. This change might cause potential scar tissue or fibrosis problems to spinal cord after laminectomy.

Therefore, a device for keeping the spinal cord protected as the laminae could help to prevent the abovementioned problems. The device is to instead the removed lamina after laminectomy. To meet the different vertebra size and different laminectomy situation, the protective device is better to be length-adjustable.

To achieve the adjustable characteristic, the protective device is designed with two or more components assembled. For medical device like this, the stability is most important. Hence, to design a protective device both stable and adjustable is an important issue nowadays.

SUMMARY

To resolve the drawbacks of the prior arts, the present invention discloses a spinal lamina protector which includes a first protection part and a second protection part. The second protection part is detachably connected with the first protection part.

Furthermore, the first protection part comprises a top plate, a first engaging portion, a first holding portion, a supporting portion and a first fixing portion. The first engaging portion is configured on the top plate, and the first holding portion is configured on the top plate, too. Thereafter, the supporting portion is configured beside the top plate, and the first fixing portion is connected with the top plate and the supporting portion simultaneously. Specifically, the supporting portion and the first holding portion form an engaging rail.

On the other hand, the second protection part comprises a lower plate, an overlapping portion, a second engaging portion, a second holding portion and a second fixing portion. The overlapping portion is configured beside the lower plate. The second engaging portion is configured on the overlapping portion, and the second holding portion is configured on the overlapping portion, too. Moreover, the second fixing portion is connected with the lower plate and the overlapping portion simultaneously.

In present invention, the second engaging portion firstly engages with the first engaging portion, making the second engaging portion insert into the engaging rail, and the second engaging portion slides in the engaging rail until the first holding portion contacts the second holding portion.

Embodiments of the invention are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like reference numerals refer to similar elements.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic drawing of the embodiment of the present invention.

FIG. **2** is a schematic drawing of the first protection part of the embodiment of the present invention.

FIG. **3** is a schematic drawing of the second protection part of the embodiment of the present invention.

FIG. **4** is a schematic drawing of latching mechanism of the embodiment of the present invention.

FIG. **5** is the other schematic drawing of latching mechanism of the embodiment of the present invention.

FIG. **6** is a schematic drawing of engaging rail of the embodiment of the present invention.

FIG. **7** is a top view drawing of FIG. **6**.

FIG. **8** is a schematic drawing illustrating how the first holding portion and the second holding portion work of the embodiment of the present invention.

FIG. **9** is a schematic drawing of the first fixing portion and the second fixing portion of the embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

In order to understand the technical features and practical efficacy of the present invention and to implement it in accordance with the contents of the specification, hereinafter, preferred embodiments of the present invention will be described in detail with reference to the accompanying drawings.

Please refer to FIG. **1**, FIG. **2** and FIG. **3** simultaneously. FIG. **1** is a schematic drawing of the embodiment of the present invention. FIG. **2** is a schematic drawing of the first protection part of the embodiment of the present invention. FIG. **3** is a schematic drawing of the second protection part of the embodiment of the present invention.

First of all, FIG. **1** illustrates the spinal lamina protector **10** of the present embodiment. Specifically, the spinal lamina protector **10** of the present embodiment is arch shaped, and the spinal lamina protector **10** comprises a first protection part **100** and a second protection part **200**. The second protection part **200** is detachably connected with the first protection part **100**. The spinal lamina protector **10** with specific structure may bear the pressure P thus to protect the spinal cord.

Please refer to FIG. **2**. FIG. **2** illustrates the top view and the side view of the first protection part **100** of the present embodiment. The first protection part **100** comprises top plate **101**, first engaging portion **103**, first holding portion **104**, supporting portion **102** and first fixing portion **105**. The first engaging portion **103** is configured on the top plate **101**, and the first holding portion **104** is configured on the top plate **101**, too.

On the other hand, the supporting portion **102** is configured beside the top plate **101**, and the first fixing portion **105** is connected with the top plate **101** and the supporting portion **102** simultaneously. Moreover, the supporting portion **102** and the first holding portion **104** form an engaging rail R.

In present embodiment, the number of supporting portion **102** and first engaging portions **103** are two. Two supporting portions **102** are configured on but not contact both sides of the top plate **101**. Furthermore, two first engaging portions **103** are directly configured on both ending sides of the top plate **101**. The end of the top plate **101** is designed to form an arch shaped notch, therefore to accommodate the terminal shape of the lower plate **201**. The first fixing portion **105** of the present embodiment is deposed at a lower place related to the top plate **101**, thus the first protection part **100** forms a structure which is similar to letter "Z", and so do the lower plate **201**, overlapping portion **202** and second fixing portion **205** of the second protection part **200**.

In this embodiment, two fixing pores FP are configured on the first fixing portion **105**, and so does the second fixing portion **205**. The number of the fixing pores FP may increase the adjustment ability of the first fixing portion **105** and second fixing portion **205**.

Please refer to FIG. 3. FIG. **3** illustrates the top view and the side view of the second protection part **200** of the present embodiment. the second protection part **200** of the present embodiment comprises lower plate **201**, overlapping portion **202**, a second engaging portion **203**, a second holding portion **204** and a second fixing portion **205**. The overlapping portion **202** is configured beside the lower plate **201**. The second engaging portion **203** is configured on the overlapping portion **202**, and the second holding portion **204** is configured on the overlapping portion, too. Moreover, the second fixing portion **205** is connected with the lower plate **202** and the overlapping portion **202** simultaneously.

In this embodiment, the number of the overlapping portion **202** and second engaging portion **203** are two. The overlapping portions **202** are configured on but not contact the lower plate **201**. On the other hand, two second engaging portions **203** are respectively configured on the end of overlapping portion **202**. The purpose of the configuration of the overlapping portion **202** is to overlap the supporting portion **102** illustrated in FIG. **2**, and the second engaging portion **203** is designed to be inserted and slides in the aforementioned engaging rail R therein.

In other words, the second engaging portion **203** firstly engages with the first engaging portion **103**, making the second engaging portion **203** insert into the engaging rail R, and the second engaging portion **203** slides in the engaging rail R until the first holding portion **104** contacts the second holding portion **204**.

In light of the specific way of creating connection between the first protection part **100** and a second protection part **200**, please refer to FIG. **4** to FIG. **7** simultaneously. FIG. **4** is a schematic drawing of latching mechanism of the embodiment of the present invention. FIG. **5** is the other schematic drawing of latching mechanism of the embodiment of the present invention. FIG. **6** is a schematic drawing of engaging rail of the embodiment of the present invention, and FIG. **7** is a top view drawing of FIG. **6**.

As shown in FIG. **4**, this figure obviously shows that the first engaging portion **103** of the present embodiment comprises a first rib **1031** and a first notch **1032**, and the second engaging portion **203** comprises a second rib **2031** and a second notch **2032**. In this embodiment, the first rib **1031** and first notch **1032** form an L-shaped structure and so do second rib **2031** and second notch **2032**. When the first engaging portion **103** has been ready for engaging with the second engaging portion **203**, the second rib **2031** may be aligned with the first notch **1032** at first.

Thereinafter, the second rib **2031** contacts and goes through the first notch **1032** along with the direction of dotted line arrow A**1**. In FIG. **4**, the aforementioned "goes through" is like a latch. In other words, pair of the overlapping portion **202** may be transiently bended in different direction and the distance therebetween two overlapping portions **202** may be enlarged while the second rib **2031** is going through the first notch **1032**, until the first notch **1032** contacts the second notch **2032**.

Please refer to FIG. **5**, after the first notch **1032** contacts the second notch **2032**, the second protection part **200** will be rotated as the directions shown by arrow A**2** and arrow A**3**. In this step, the purpose of the present embodiment is to make the second rib **2031** insert into the engaging rail R. Please notice that the end point of first holding portion **104** needs to be designed on the route of second rib **2031** while the second rib **2031** is rotated along with the direction of arrow A**2**. This mechanism may create a transient stopper which is played by first holding portion **104** and make the first holding portion **104** successfully holds the second rib **2031** after the second rib **2031** has been inserted into the engaging rail R.

After the second rib **2031** has been inserted into the engaging rail R, as shown in FIG. **6**, the first rib **1031** will be inserted into the engaging rail R, too. Otherwise, the first rib **1031** and the second rib **2031** are able to slide in the engaging rail R, thus the first protection part **100** will slides along the direction of dotted line arrow A**5** and the second protection part **200** will slides along the direction of arrow A**4**.

In FIG. **6**, the dotted line arrow A**5** and the arrow A**4** further show how to adjust the length of the spinal lamina protector **10**. That is, the first holding portion **104** and the second holding portion **204** linearly moves in the same line and so do the first engaging portion **103** and the second engaging portion **203**.

When the spinal lamina protector **10** extends the length to the extreme limit per se, the first holding portion **104** contact against the second holding portion **204**, and the L-shaped first rib **1031** and first notch **1032** contact against the L-shaped seconds rib **2031** and second notch **2032**, too. On the other hand, the first holding portion **104** always holds the seconds rib **2031** and second notch **2032** from the top, and the second holding portion **204** always holds the first rib **1031** and first notch **1032** from the top, too.

Moreover, to more clearly point out the mechanism which works in the present embodiment, the second protection part **200** is shaded in FIG. **7**. Therefore, FIG. **7** further illustrates that the top plate **101** slides on the upper surface of lower plate **201**, and the overlapping portion **202** slides on the upper surface of supporting portion **102** in the present embodiment.

The abovementioned mechanism may not only design the adjustable length range of the spinal lamina protector **10**, but also make the spinal lamina protector **10** become more stable. Please refer to FIG. **8**, FIG. **8** is a schematic drawing illustrating how the first holding portion and the second holding portion work of the embodiment of the present invention.

As shown in FIG. **7** and FIG. **8**, the reason why the spinal lamina protector **10** of the present embodiment becomes more stable is that the first holding portion **104** restricts the rotation of the second engaging portion **203** from the top and the second holding portion **204** restricts the rotation of the first engaging portion **103** while the first engaging portion **103** and the second engaging portion **203** are inserting into the engaging rail R. Therefore, the first engaging portion **103**, the second engaging portion **203**, the first holding portion **104**, and the second holding portion **204** form double levers against each other, creating a specific and stable structure of spinal lamina protector **10** for bearing the pressure P.

Please refer to FIG. **9**, FIG. **9** is a schematic drawing of the first fixing portion and the second fixing portion of the embodiment of the present invention. FIG. **9** shows the installation method of the spinal lamina protector **10** of the present embodiment. Fasteners such as nails or pins may be used for fixing the length adjusted spinal lamina protector **10** via the fixing pore FP along with the direction of arrow A**6** on the vertebra after laminectomy, thus to form the above-mentioned double lever structure.

As is understood by a person skilled in the art, the foregoing preferred embodiments of the present invention are illustrated of the present invention rather than limiting of the present invention. It is intended to cover various modifications and similar arrangements included within the spirit and scope of the appended claims, the scope of which should be accorded the broadest interpretation so as to encompass all such modifications and similar structure. While the preferred embodiment of the invention has been illustrated and described, it will be appreciated that various changes can be made therein without departing from the spirit and scope of the invention.

What is claimed:

1. A spinal lamina protector, comprising:

a first protection part, comprising:

a top plate;

a first engaging portion, configured on the top plate;

a first holding portion, configured on the top plate;

a supporting portion, configured beside the top plate;

a first fixing portion, connected with the top plate and the supporting portion simultaneously;

wherein the supporting portion and the first holding portion form an engaging rail;

a second protection part, detachably connected with the first protection part, wherein the second protection part comprises:

a lower plate;

an overlapping portion, configured beside the lower plate;

a second engaging portion, configured on the overlapping portion;

a second holding portion, configured on the overlapping portion;

a second fixing portion, connected with the lower plate and the overlapping portion simultaneously;

wherein the second engaging portion firstly engages with the first engaging portion, making the second engaging portion insert into the engaging rail, and the second engaging portion slides in the engaging rail until the first holding portion contacts the second holding portion.

2. The spinal lamina protector as claimed in claim **1**, wherein the first engaging portion further comprises a first rib and a first notch.

3. The spinal lamina protector as claimed in claim **1**, wherein the second engaging portion further comprises a second rib and a second notch.

4. The spinal lamina protector as claimed in claim **2**, wherein the second holding portion restricts rotation of the first engaging portion while first engaging portion is inserting into the engaging rail.

5. The spinal lamina protector as claimed in claim **3**, wherein the first holding portion restricts rotation of the second engaging portion while the second engaging portion is inserting into the engaging rail.

6. The spinal lamina protector as claimed in claim **1**, wherein the top plate slides on the lower plate.

7. The spinal lamina protector as claimed in claim **1**, wherein the overlapping portion slides on the supporting portion.

8. The spinal lamina protector as claimed in claim **1**, wherein the spinal lamina protector is arch shaped.

9. The spinal lamina protector as claimed in claim **1**, wherein the first fixing portion further comprises at least one fixing pore.

10. The spinal lamina protector as claimed in claim **1**, wherein the second fixing portion further comprises at least one fixing pore.

* * * * *